

Fundamental of Linear Algebra and Statistics

(25AS102) (Evening Shift)

①

1. (a)

x	f	$d = x - a$ $= x - 20$	fd
5	8	-15	-120
10	15	-10	-150
15	20	-5	-100
20	32	0	0
25	23	5	115
30	17	10	170
35	05	15	75
$\Sigma f = 120$			$\Sigma fd = -10$

②

$$\bar{x} = a + \frac{\Sigma fd}{\Sigma f} = 20 - \frac{10}{120} = 20 - \frac{1}{12} = 19.916$$

(b)

wages in thousands frequency (No. of workers)

Below 100

8

100 - 200

⑫ f_1

200 - 300

⑫⑤ f_m

300 - 400

⑫⑤ f_2

400 - 500

10

Above 500

6

①.5

Highest frequency = 25, Modal class = 200-300

$f_m = 25$, $f_1 = 12$, $f_2 = 15$, $h = 100$, $L = 200$

$$\text{Mode} = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

$$= 200 + \frac{25 - 12}{50 - 12 - 15} \times 100 = 200 + \frac{13}{23} \times 100$$

$$= 256.52$$

2. (a)

Q:-2 @ class interval

class interval	x (midpoint)	f	fx	x^2	fx^2 ②
10 - 20	15	1	15	225	225
20 - 30	25	20	500	625	12,500
30 - 40	35	69	2415	1225	84,525
40 - 50	45	108	4860	2025	218,700
50 - 60	55	78	4290	3025	235,950
60 - 70	65	22	1430	4225	92,950
70 - 80	75	2	150	5625	11,250
		$\Sigma f = 300$	$\Sigma fx = 13,660$		$\Sigma fx^2 = 656,100$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{13,660}{300} = 45.53$$

$$S.D = \sigma = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$$

$$= \sqrt{\frac{656,100}{300} - \left(\frac{13,660}{300}\right)^2} = 10.68$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{10.68}{45.53} \times 100 = 23.47$$

⑥ Mean = $\bar{x} = \frac{120 + 125 + 118 + 130 + 122 + 127}{6}$

$$= 123.67$$

Mean deviation (M.D) = $|120 - 123.67| + |125 - 123.67| + |118 - 123.67| + |130 - 123.67| + |122 - 123.67| + |127 - 123.67|$

$$= \frac{3.67 + 1.33 + 5.67 + 6.33 + 1.67 + 3.33}{6}$$

$$= \frac{22}{6} = 3.67$$

coefficient of mean deviation = $\frac{M.D}{\bar{x}} = \frac{3.67}{123.67}$

$$= 0.029$$

(3)
(2)

x	y	x^2	x^3	x^4	xy	x^2y
2	14	4	8	16	28	56
3	17	9	27	81	51	153
4	20	16	64	256	80	320
5	24	25	125	625	120	600
6	29	36	216	1296	174	1044
<u>20</u>	<u>104</u>	<u>90</u>	<u>440</u>	<u>2274</u>	<u>453</u>	<u>2173</u>

(3)

Let $y = a + bx + cx^2$ ——— (1) ——— 0.5

The normal equations are

$$\begin{aligned} \sum y &= na + b\sum x + c\sum x^2 \\ \sum xy &= a\sum x + b\sum x^2 + c\sum x^3 \\ \sum x^2y &= a\sum x^2 + b\sum x^3 + c\sum x^4 \end{aligned}$$

(1)

$\Rightarrow 104 = 5a + 20b + 90c$
 $453 = 20a + 90b + 440c$
 $2173 = 90a + 440b + 2274c$

on solving these eqns,

$a = 11, b = 0.844, c = 0.357$

$y = 11 + 0.844x + 0.357x^2$ ——— 0.5

(b) (i) $y = ax + \frac{b}{x}$
 Normal equations are

$$\begin{aligned} \sum xy &= a\sum x^2 + b\sum \frac{1}{x} \\ \sum \frac{y}{x} &= na + b\sum \frac{1}{x^2} \end{aligned}$$

(1)

(ii) $y = \frac{a}{\sqrt{x}} + bx^{3/2}$

$$\begin{aligned} \sum \frac{y}{\sqrt{x}} &= a\sum \frac{1}{x} + b\sum x^{3/2} \\ \sum x^2y &= a\sum x^{3/2} + b\sum x^4 \end{aligned}$$

(iii) $y = a_1e^{3t} + a_2e^{-4t}$

$$\begin{aligned} \sum ye^{3t} &= a_1\sum e^{6t} + a_2\sum e^{-8t} \\ \sum ye^{-4t} &= a_1\sum e^{-t} + a_2\sum e^{-8t} \end{aligned}$$

Q4 a) Rank correlation

4

$$r = 1 - \frac{6}{n(n^2-1)} \left[\sum d^2 + \frac{1}{12} m_1(m_1^2-1) + \frac{1}{12} m_2(m_2^2-1) + \frac{1}{12} m_3(m_3^2-1) \right]$$

0.5

x	y	R ₁	R ₂	d	d ²
23	18	10	10	0	0
27	20	9	9	0	0
28	22	7.5	7	0.5	0.25
28	27	7.5	5.5	2	4
29	21	6	8	-2	4
30	29	5	2	3	9
31	27	4	5.5	-1.5	2.25
33	29	3	2	1	1
35	28	2	4	-2	4
36	29	1	2	-1	1

1.5

$$\sum d^2 = 27.75$$

tie for value 28 at position 7, 8 $\boxed{m_1 = 2}$
final rank = $\frac{7+8}{2} = 7.5$

tie for value 29 at position 1, 2, 3 $\boxed{m_2 = 3}$
final rank = $\frac{1+2+3}{3} = 2$

0.5

tie for value 27 at position 5, 6. $\boxed{m_3 = 2}$
final rank = $\frac{5+6}{2} = 5.5$

$$r = 1 - \frac{6}{10(99)} \left[27.75 + \frac{1}{12} 2(3) + \frac{1}{12} 3(8) + \frac{1}{12} 2(3) \right]$$

$$r = 1 - 0.18618 = \boxed{0.8138}$$

0.5

Q4 b) Regression lines.

$$\begin{aligned} x + 2y - 5 &= 0 \\ 2x + 3y - 8 &= 0 \end{aligned}$$

$$\text{variance of } x = \sigma^2 x = 12$$

a) mean values of x & y . $(\bar{x}, \bar{y})$ is common point of intersection of regression lines

$$\begin{aligned} \bar{x} + 2\bar{y} &= 5 \\ 2\bar{x} + 3\bar{y} &= 8 \end{aligned}$$

$$\Rightarrow \boxed{\bar{x}=1, \bar{y}=2}$$

(1)

(5)

b) Coefficient of correlation between x & y .

$$r = \pm \sqrt{b_{yx} \times b_{xy}}$$

Let $x+2y-5=0$ is regression line of y on x

$$\therefore y = \frac{-x+5}{2}$$

$$\boxed{b_{yx} = -\frac{1}{2}}$$

Let $2x+3y-8=0$ is regression line of x on y

$$x = \frac{-3y+8}{2}$$

$$\boxed{b_{xy} = -\frac{3}{2}}$$

$$r = \sqrt{\frac{-1}{2} \times \frac{-3}{2}} = \frac{\sqrt{3}}{2} \quad \because b_{yx} \& b_{xy} < 0$$

$$\therefore \boxed{r = -\frac{\sqrt{3}}{2}} \quad \text{--- (1)}$$

c) Standard deviation of y σ_y

$$\therefore b_{yx} = r \frac{\sigma_y}{\sigma_x}$$

$$-\frac{1}{2} = \frac{-\sqrt{3} \times \sigma_y}{\frac{1}{\sqrt{12}}} \Rightarrow +\frac{1}{2} = \frac{+\sqrt{3}}{2} \frac{\sigma_y}{\sqrt{3}} \Rightarrow \boxed{\sigma_y = 2} \quad \text{--- (1)}$$

Section-B.

Q.5 a) Measure of skewness $\gamma_1 = \frac{\mu_3}{\sqrt{\mu_2^3}}$

(0.5)

value x	Freq. f	$f(x-\bar{x})$	$f(x-\bar{x})^2$	$f(x-\bar{x})^3$	$f(x-\bar{x})^4$	fx	
6	4	-75.2727	1416.4958	-26655.87	501615.13	24	
12	7	-89.7273	1150.1404	-14742.709	188974.73	84	
18	9	-61.3636	418.388	-2852.1483	19449.875	162	
24	18	-14.7273	12.0495	-9.85875	8.06625	432	
30	15	77.7273	402.7685	2087.073	10814.836	450	
36	10	111.818	1250.330	13980.9692	156332.45	360	
42	3	51.5455	885.644	15416.9849	261455.469	126	
Σ	66	0.0000	5535.818	-12976.066	1138650.775	1638	(2)

$$\mu_1 = 0$$

$$\mu_2 = \frac{\sum f(x-\bar{x})^2}{\sum f}$$

$$\mu_3 = \frac{\sum f(x-\bar{x})^3}{\sum f}$$

(6)

$$\mu_2 = 83.87603$$

$$\mu_3 = -196.60706$$

(1)

$$\gamma_1 = -0.2559$$

-vely skewed distribution.

(0.5)

Q.5 b) $A = 2, \mu_1' = 1, \mu_2' = 2.5, \mu_3' = 5.5, \mu_4' = 16.$

Moments about mean

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - (\mu_1')^2 = 2.5 - (1)^2 = 1.5$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3 = 0$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4 = 6$$

Moments about origin

$$v_1 = x = A + \mu_1' = \boxed{v_1 = 3}$$

$$v_2 = \mu_2 + \bar{x}^2 \Rightarrow \boxed{v_2 = 10.5}$$

$$v_3 = \mu_3 + 3\mu_2\bar{x} + \bar{x}^3 = 40.5$$

$$\boxed{v_3 = 40.5}$$

$$v_4 = \mu_4 + 4\mu_3\bar{x} + 6\mu_2\bar{x}^2 + \bar{x}^4 = \boxed{168 = v_4}$$

(2)

Q.6 a) $y = ab^x$

taking $\log_{10} y = \log_{10} a + x \log_{10} b$

$$y = A + Bx$$

$$\log_{10} y = Y$$

$$\log_{10} a = A$$

$$\log_{10} b = B$$

Normal eqⁿs are

$$\sum Y = nA + B \sum x$$

$$\sum xY = A \sum x + B \sum x^2$$

(1)

x	y	$Y = \log_{10} y$	x^2	xY
2	8.3	0.91907	4	1.8382
3	15.4	1.18752	9	3.5616
4	33.1	1.51982	16	6.0792
5	65.2	1.81424	25	9.0710
6	127.4	2.10516	36	12.6312
Σ	20	7.54581	90	33.1812

$$\boxed{n = 5}$$

(2)

$$A = 0.309602, B = 0.29989$$

$$a = \text{antilog}_{10} A = 2.0396,$$

$$b = \text{antilog}_{10} B = 1.99475$$

$$\therefore y = (2.0398) (1.99475)^x \quad \text{--- (1)} \quad \textcircled{7}$$

Q.6 b) Regression line of x on y .

$$(x - \bar{x}) = b_{xy} (y - \bar{y})$$

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2} \quad \text{--- (1)} \quad \textcircled{11}$$

x	y	y^2	xy
1	8	64	8
3	12	144	36
5	15	225	75
7	17	289	119
8	18	324	144
10	20	400	200
Σ	34	90	1446
			582

$$\bar{x} = \frac{\sum x}{n} = 5.6667 \quad \text{--- (1)} \quad \textcircled{1}$$

$$\bar{y} = \frac{\sum y}{n} = 15.$$

$$b_{xy} = \frac{432}{576} = 0.75$$

$$b_{xy} = 0.75$$

$$x = 0.75y + 5.58333 \quad \text{--- (1)} \quad \textcircled{1}$$